

Comparison of AWS Services and Features

ECS Network Mode Comparison

Amazon Elastic Container Service (ECS) allows you to run Docker-based containers on the cloud. Amazon ECS has two launch types for operation: EC2 and Fargate. The EC2 launch type provides EC2 instances as hosts for your Docker containers. For the Fargate launch type, AWS manages the underlying hosts so you can focus on managing your containers instead. The details and configuration on how you want to run your containers are defined on the [ECS Task Definition](#) which includes options on networking mode.

In this post, we'll talk about the different networking modes supported by Amazon ECS and determine which mode to use for your given requirements.

ECS Network Modes

Amazon Elastic Container Service supports four networking modes: **Bridge**, **Host**, **awsvpc**, and **None**. This selection will be set as the Docker networking mode used by the containers on your ECS tasks.

Configure task and container definitions

A task definition specifies which containers are included in your task and how they interact with each other. You can also specify data volumes for your containers to use. [Learn more](#)

Task Definition Name* ⓘ

Requires Compatibilities* EC2

Task Role ⓘ
Optional IAM role that tasks can use to make API requests to authorized AWS services. Create an Amazon Elastic Container Service Task Role in the [IAM Console](#) ⓘ

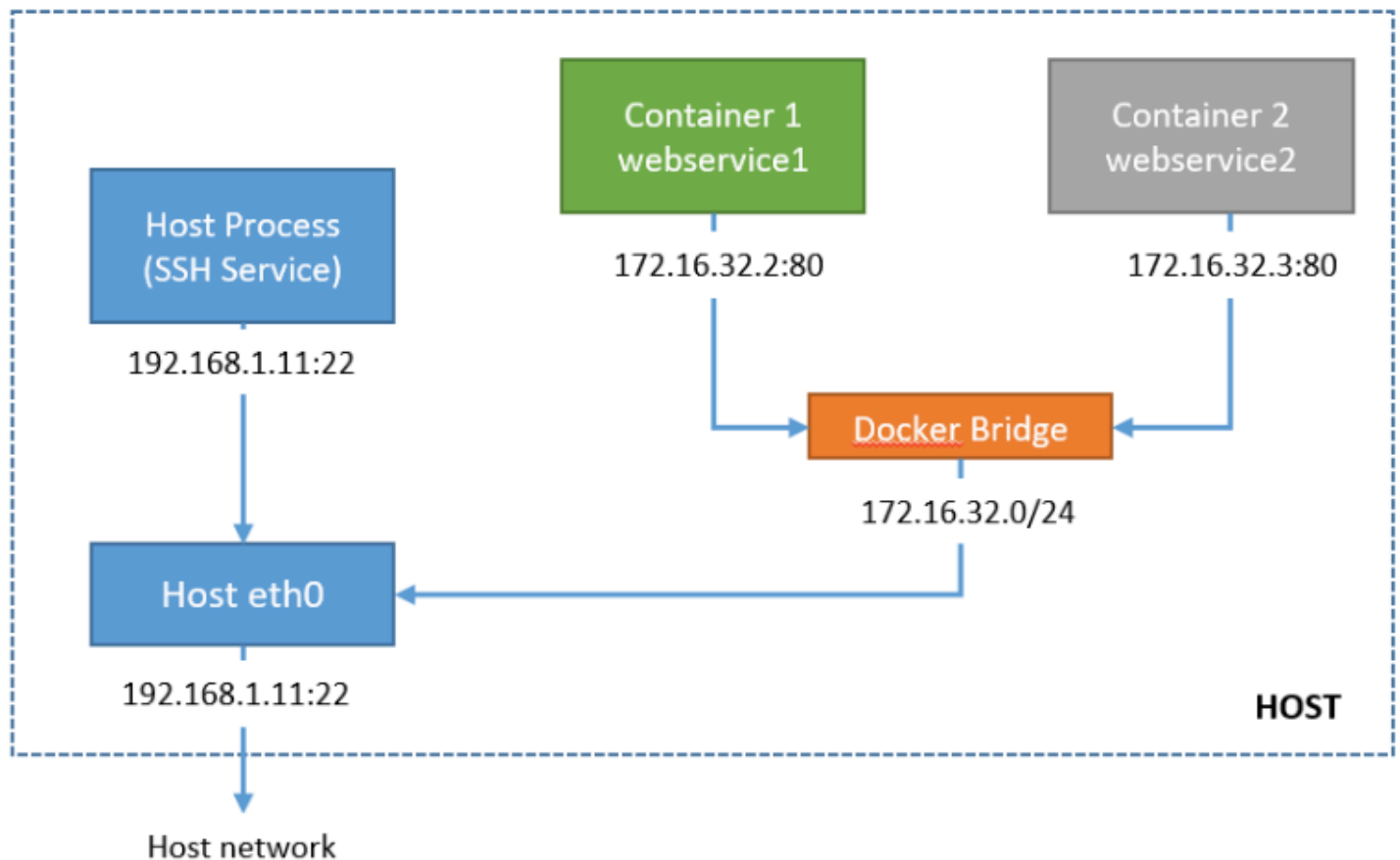
Network Mode ⓘ

 Bridge
 Host
 awsvpc
 None

Bridge network mode - Default

When you select the **<default>** network mode, you are selecting the **Bridge** network mode. This is the default mode for Linux containers. For Windows Docker containers, the **<default>** network mode is **NAT**. You must select **<default>** if you are going to register task definitions with Windows containers.

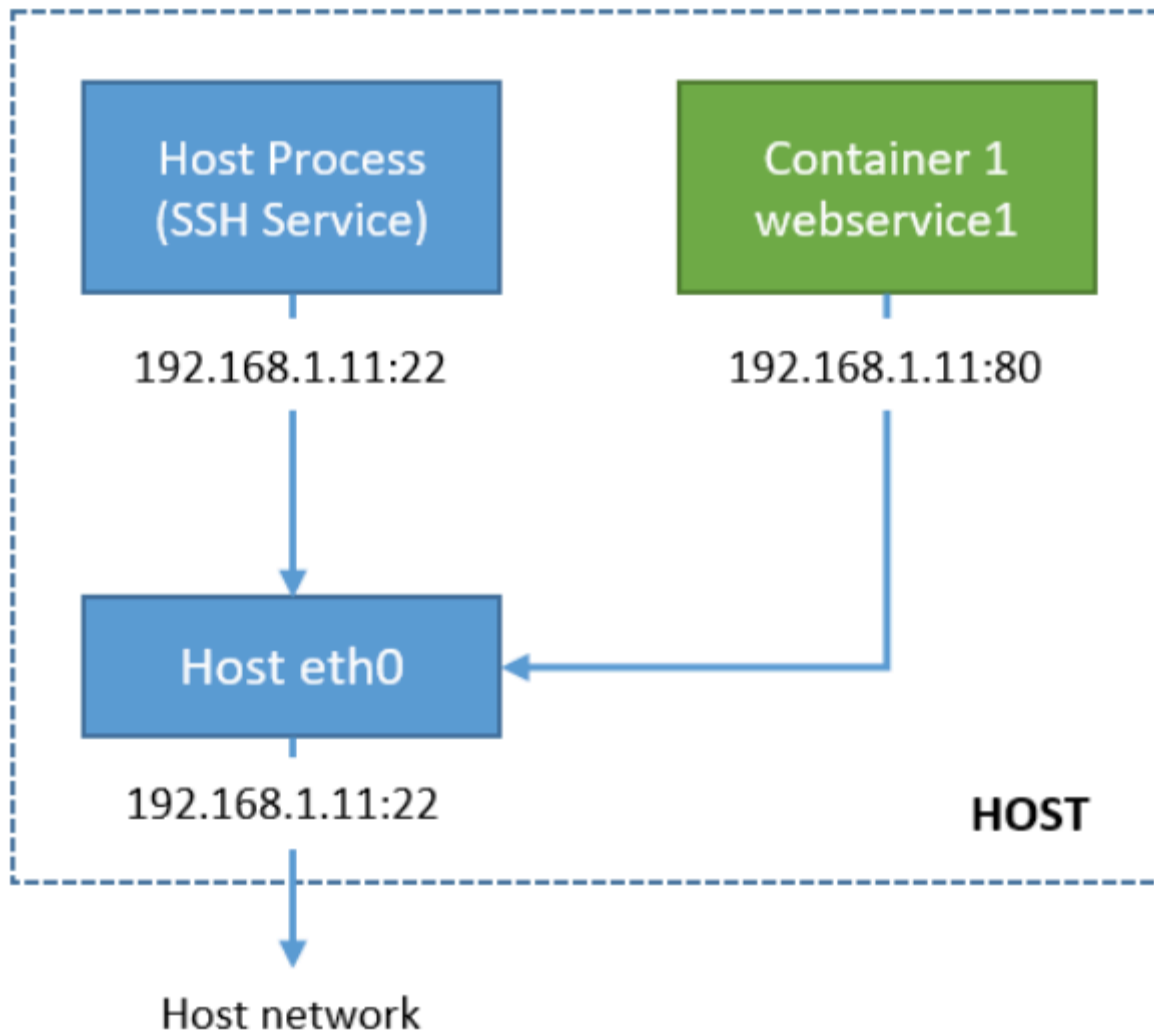
Bridge network mode utilizes Docker's built-in virtual network which runs inside each container. A bridge network is an internal network namespace in the host that allows all containers connected on the same bridge network to communicate. It provides isolation from other containers not connected to that bridge network. The Docker driver handles this isolation on the host machine so that containers on different bridge networks cannot communicate with each other.



This mode can take advantage of dynamic host port mappings as it allows you to run the same port (ex: port 80) on each container, and then map each container port to a different port on the host. However, this mode does not provide the best networking performance because the bridge network is virtualized and Docker software handles the traffic translations on traffic going in and out of the host.

Host network mode

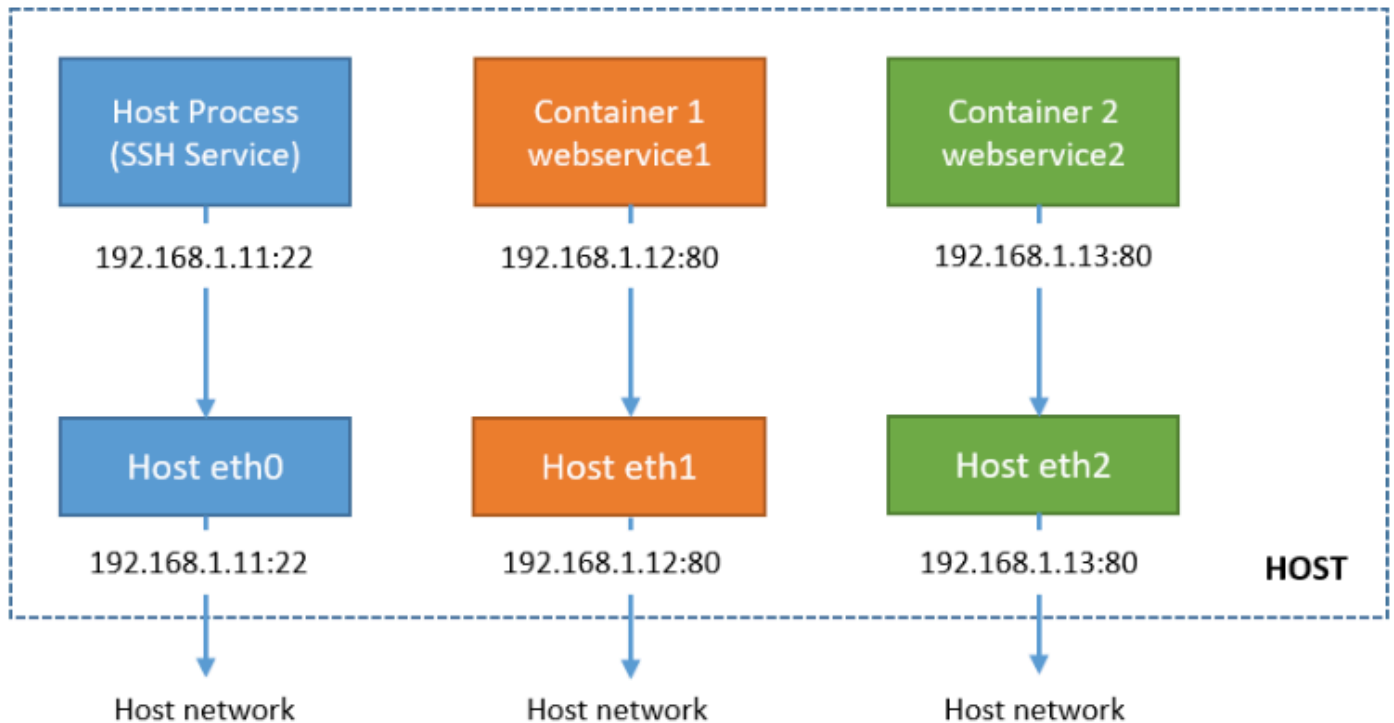
Host network mode bypasses the Docker's built-in virtual network and maps container ports directly to your EC2 instance's network interface. This mode shares the same network namespace of the host EC2 instance so your containers share the same IP with your host IP address. This also means that you can't have multiple containers on the host using the same port. A port used by one container on the host cannot be used by another container as this will cause conflict.



This mode offers faster performance than the bridge network mode since it uses the EC2 network stack instead of the virtual Docker network.

awsvpc mode

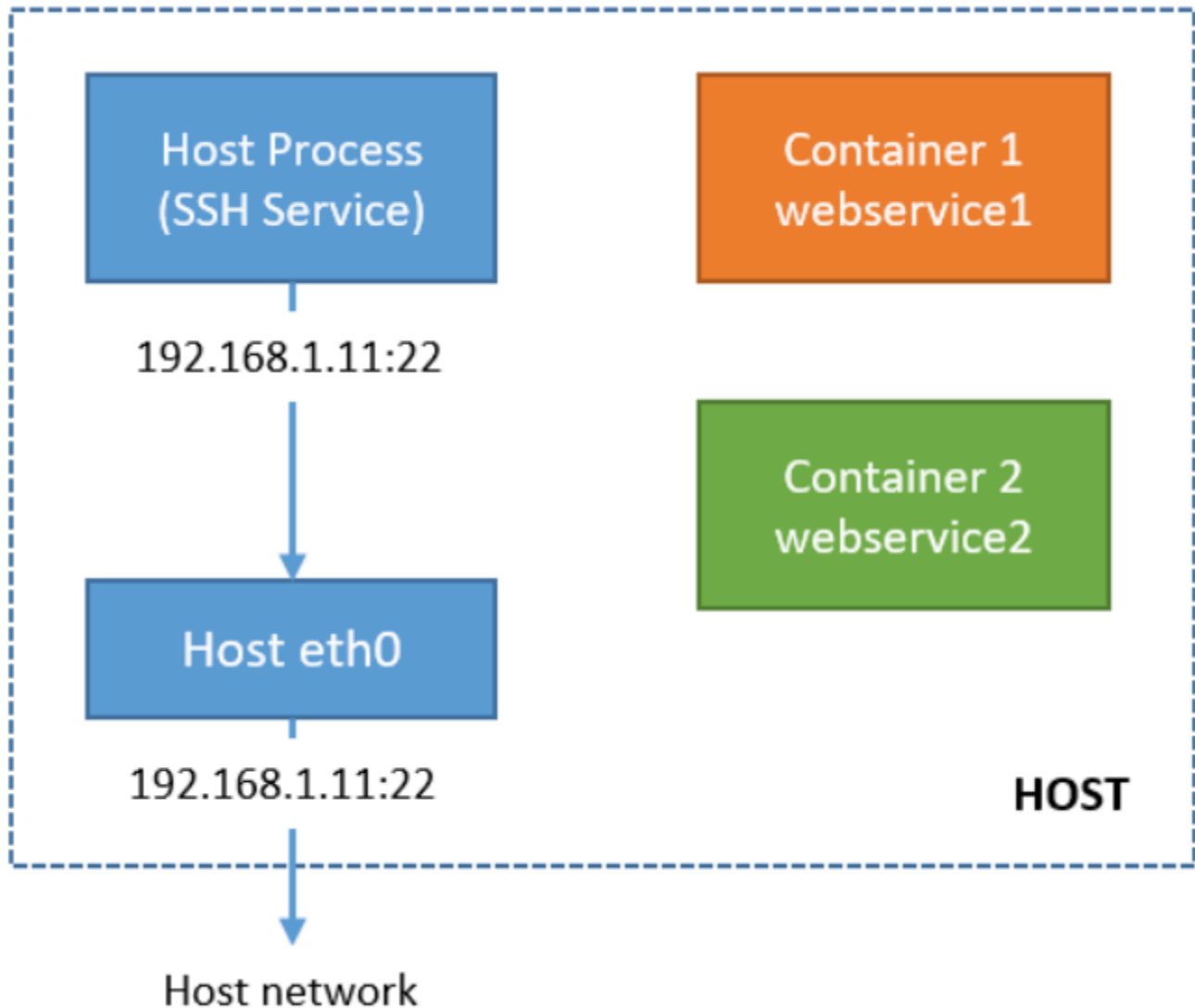
The **awsvpc** mode provides an elastic network interface for each task definition. If you have one container per task definition, each container will have its own elastic network interface and will get its own IP address from your VPC subnet IP address pool. This offers faster performance than the bridge network since it uses the EC2 network stack, too. This essentially makes each task act like their own EC2 instance within the VPC with their own ENI, even though the tasks actually reside on an EC2 host.



Awsvpc mode is recommended if your cluster will contain several tasks and containers as each can communicate with their own network interface. This is the only supported mode by the ECS Fargate service. Since you don't manage any EC2 hosts on ECS Fargate, you can only use awsvpc network mode so that each task gets its own network interface and IP address.

None network mode

This mode completely disables the networking stack inside the ECS task. The loopback network interface is the only one present inside each container since the loopback interface is essential for Linux operations. You can't specify port mappings on this mode as the containers do not have external connectivity.



You can use this mode if you don't want your containers to access the host network, or if you want to use a custom network driver other than the built-in driver from Docker. You can only access the container from inside the EC2 host with the Docker command.

References:

https://docs.aws.amazon.com/AmazonECS/latest/developerguide/task_definition_parameters.html#network_mode
<https://docs.aws.amazon.com/AmazonECS/latest/developerguide/task-networking.html>
<https://docs.aws.amazon.com/AmazonECS/latest/userguide/fargate-task-networking.html>



Application Load Balancer vs Network Load Balancer vs Gateway Load Balancer

FEATURE	Application Load Balancer	Network Load Balancer	Gateway Load Balancer
Protocols	HTTP, HTTPS, gRPC	TCP, UDP, TLS	IP
Platforms	VPC	VPC	VPC
Health checks	HTTP, HTTPS, gRPC...	TCP, HTTP, HTTPS	TCP, HTTP, HTTPS
Cloudwatch Metrics	Yes	Yes	Yes
Logging	Yes	Yes	Yes
Zonal Failover	Yes	Yes	Yes
Connection Draining (deregistration delay)	Yes	Yes	Yes
Load Balancing to multiple ports on the same instance	Yes	Yes	Yes
IP addresses as targets	Yes	Yes (TCP, TLS)	Yes
Load Balancer deletion protection	Yes		
Configuration idle connection timeout	Yes		
Cross-zone load balancing	Yes	Yes	Yes
Sticky sessions	Yes	Yes	Yes
Static IP		Yes	
Elastic IP address		Yes	
Preserve Source IP address	Yes	Yes	Yes
Resource-based IAM permissions	Yes	Yes	Yes
Slow start	Yes		
Web sockets	Yes	Yes	Yes



PrivateLink Support		Yes (TCP, TLS)	Yes (GWLBE)
Source IP address CIDR-based routing	Yes		
Layer 7			
Path-based routing	Yes		
Host-based routing	Yes		
Native HTTP/2	Yes		
Redirects	Yes		
Fixed response	Yes		
Lambda Functions as targets	Yes		
HTTP header-based routing	Yes		
HTTP method-based routing	Yes		
Query string parameter-based routing	Yes		
Security			
SSL offloading	Yes	Yes	
Server Name Indication (SNI)	Yes	Yes	
Back-end server encryption	Yes	Yes	
User authentication	Yes		
Session Resumption	Yes	Yes	
Terminates flow/ proxy behavior	Yes	Yes	Yes



Common features between the load balancers:

- Has instance health check features
- Has built-in CloudWatch monitoring
- Logging features
- Support zonal failover
- Support cross-zone load balancing (evenly distributes traffic across registered instances in enabled AZs)
- Resource-based IAM permission policies
- Tag-based IAM permissions
- Flow stickiness - all packets are sent to one target and return the traffic that comes from the same target.



S3 Pre-Signed URLs vs CloudFront Signed URLs vs Origin Access Control

S3 Pre-signed URLs	CloudFront Signed URLs	Origin Access Control (OAC)
All S3 buckets and objects, by default, are private . Only the object owner has permission to access these objects. Pre-signed URLs use the owner's security credentials to grant others time-limited permission to download or upload objects.	You can control user access to your private content in two ways - Restrict access to files in CloudFront edge caches - Restrict access to files in your Amazon S3 bucket (unless you've configured it as a website endpoint).	You can configure an S3 bucket as the origin of a CloudFront distribution. OAC prevents users from viewing your S3 files by simply using the direct URL for the file. Instead, they would need to access it through a CloudFront URL.
When creating a pre-signed URL, you (as the owner) need to provide the following: - Your security credentials - An S3 bucket name - An object key - Specify the HTTP method (GET to download the object or PUT to upload an object) - Expiration date and time of the URL.	You can configure CloudFront to require that users access your files using either signed URLs or signed cookies . You then develop your application either to create and distribute signed URLs to authenticated users or to send Set-Cookie headers that set signed cookies on the viewers for authenticated users. When you create signed URLs or signed cookies to control access to your files, you can specify the following restrictions: - An expiration date and time for the URL (Optional) The date and time the URL becomes valid (Optional) The IP address or range of addresses of the computers that can be used to access your content You can use signed URLs or signed cookies for any CloudFront distribution, regardless of whether the origin is an Amazon S3 bucket or an HTTP server.	To require that users access your content through CloudFront URLs, you perform the following tasks: - Create a special CloudFront user called an origin access control. - Give the origin access control permission to read the files in your bucket. - Remove permission for anyone else to use Amazon S3 URLs to read the files (through bucket policies or ACLs). You cannot set OAC if your S3 bucket is configured as a website endpoint.



S3 Transfer Acceleration vs Direct Connect vs VPN

S3 Transfer Acceleration (TA)

- Amazon S3 Transfer Acceleration makes public Internet transfers to S3 faster, as it leverages Amazon CloudFront's globally distributed AWS Edge Locations.
- There is no guarantee that you will experience increased transfer speeds. If S3 Transfer Acceleration is not likely to be faster than a regular S3 transfer of the same object to the same destination AWS Region, AWS will not charge for the use of S3 TA for that transfer.
- This is not the best transfer service to use if transfer disruption is not tolerable.
- S3 TA provides the same security benefits as regular transfers to Amazon S3. This service also supports multi-part upload.
- **S3 TA vs Direct Connect**
 - AWS Direct Connect is a good choice for customers who have a private networking requirement or who have access to AWS Direct Connect exchanges. S3 Transfer Acceleration is best for submitting data from distributed client locations over the public Internet, or where variable network conditions make throughput poor.
- **S3 TA vs VPN**
 - You typically use (IPsec) VPN if you want your resources contained in a private network. VPN tools such as OpenVPN allow you to set up stricter access controls if you have a private S3 bucket. You can complement this further with the increased speeds from S3 TA.

AWS Direct Connect

- Using AWS Direct Connect, data that would have previously been transported over the Internet can now be delivered through a **private physical network connection** between AWS and your datacenter or corporate network. Customers' traffic will remain in AWS global network backbone, after it enters AWS global network backbone.
- Benefits of Direct Connect vs internet-based connections
 - reduced costs
 - increased bandwidth
 - a more consistent network experience
- Each AWS Direct Connect connection can be configured with one or more **virtual interfaces**. Virtual interfaces may be configured to access AWS services such as Amazon EC2 and Amazon S3 using public IP space, or resources in a VPC using private IP space.
- You can run IPv4 and IPv6 on the same virtual interface.
- Direct Connect does not support multicast.
- A Direct Connect connection is **not redundant**. Therefore, a second line needs to be established if redundancy is required. Enable *Bidirectional Forwarding Detection* (BFD) when configuring your connections to ensure fast detection and failover.
- AWS Direct Connect offers SLA.
- Direct Connect vs IPsec VPN
 - A VPC VPN Connection utilizes IPsec to establish **encrypted network connectivity** between your intranet and Amazon VPC **over the Internet**. VPN Connections can be configured in minutes and are a good solution if you have an immediate need, have low to modest bandwidth



requirements, and can tolerate the inherent variability in Internet-based connectivity. AWS Direct Connect **does not involve the Internet**; instead, it uses **dedicated, private network connections** between your intranet and Amazon VPC.

- You can combine one or more Direct Connect dedicated network connections with the Amazon VPC VPN. This combination provides an IPsec-encrypted private connection that also includes the benefits of Direct Connect.

AWS VPN

- AWS VPN is comprised of two services:
 - AWS Site-to-Site VPN enables you to securely connect your on-premises network or branch office site to your Amazon VPC.
 - AWS Client VPN enables you to securely connect users to AWS or on-premises networks.
- Data transferred between your VPC and datacenter routes over an encrypted VPN connection to help maintain the confidentiality and integrity of data in transit.
- If data that passes through Direct Connect moves in a dedicated private network line, AWS VPN instead encrypts the data before passing it through the Internet.
- VPN connection throughput can depend on multiple factors, such as the capability of your customer gateway, the capacity of your connection, average packet size, the protocol being used, TCP vs. UDP, and the network latency between your customer gateway and the virtual private gateway.
- All the VPN sessions are **full-tunnel VPN**. (cannot split tunnel)
- AWS Site-to-Site VPN enables you to create **failover** and CloudHub solutions **with AWS Direct Connect**.
- AWS Client VPN is designed to connect devices to your applications. It allows you to choose from an **OpenVPN-based client**.



Backup and Restore vs Pilot Light vs Warm Standby vs Multi-site

Backup and Restore	Pilot Light
This DR plan provides the slowest system restoration after a DR event.	The pilot light method gives you a quicker recovery time than the backup-and-restore method because the core pieces of the system are already running and are continually kept up to date, but is not as fast as Warm Standby.
You take frequent snapshots of your data such as those in Amazon EBS Volumes and Amazon RDS databases, and you store them in a durable and secure storage location such as Amazon S3.	You can maintain a pilot light by configuring and running the most critical core elements of your system in AWS. When the time comes for recovery, you can rapidly provision a full-scale production environment around the critical core.
- There are many ways for you to move data in and out of S3 - Transfer over the network via S3 Transfer Acceleration - Transfer over a dedicated network line using AWS Direct Connect	Pilot light is an example of active/passive failover configuration.
With S3 Glacier, you get to reduce a large portion of your costs compared to using S3 Standard, since Glacier is meant for long term archival storage which is perfect for backups.	Infrastructure elements for the pilot light itself typically include your database servers, which would be configured for data mirroring replication.
- AWS Storage Gateway enables snapshots of your on-premises data volumes to be transparently copied into S3 for backup. - Storage-cached volumes allow you to store your primary data in S3, but keep your frequently accessed data local for low-latency access.	Restoring the rest of the system includes utilizing EBS snapshots and EC2 AMIs that you should be regularly generating.
Gateway-VTL of AWS Storage Gateway serves as a replacement for traditional magnetic tape backup.	Pilot light tends to be more costly than backup and restore since you leave a few core AWS resources running all the time.
You can quickly create local volumes or Amazon EBS volumes from snapshots in S3.	- From a networking point of view, you have two main options for provisioning web servers: - Use Elastic IP addresses, which can be pre-allocated and pre-identified, and associate them with your instances.



	• Use Elastic Load Balancing (ELB) to distribute traffic to multiple instances. You would then update your DNS records to point at your EC2 instance or point to your load balancer using a CNAME.
• You can create AMIs out of your EC2 instances which preserve the following: • A template for the root volume for the instance (for example, an operating system, an application server, and applications) • Launch permissions that control which AWS accounts can use the AMI to launch instances • A block device mapping that specifies the volumes to attach to the instance when it's launched	• Consider redundancy especially at your data layer (enable multi-AZ, cluster sharding, etc).
• Backup and restore is used in combination with other DR plans since it is crucial to always have a working backup of your system.	• If your data is constantly changing and failover occurs, you would have to reverse replicate your data in the DR site back to the primary site, so that any data updates received while the primary site was down can be replicated back, without the loss of data.
Warm Standby	Multi-site
• This DR plan is faster in system restoration than performing Pilot Light after a DR event, but is not as fast as having a Multi-site System.	• This DR plan is the fastest in system restoration during a DR event.
• Warm standby describes a DR scenario in which a scaled-down version of a fully functional environment is always running in the cloud.	• Multi-site is a one-to-one copy of your infrastructure that is located and running in another region or AZ, known as an active-active configuration.
• Since it is not only your core elements that are running all the time, warm standby is usually more costly than pilot light.	• Because of this, multi-site is the most expensive among all DR plans.
• Warm standby is another example of active/passive failover configuration.	• Multi-site gives you the best RTO and RPO as no downtime is expected and little to no data loss should be experienced.



• Servers can be left running in a minimum number of EC2 instances on the smallest sizes possible. Once failover occurs, quickly resize them and add scaling capabilities. It is best to place these instances behind a load balancer as well.	• In addition to recovery point options, there are various replication methods, such as synchronous and asynchronous methods.
• For the data layer, the practice is similar to pilot light where a standby resource is present and changing data is constantly being replicated to the other.	• You can use a DNS service that supports weighted routing, such as Amazon Route 53, to route production traffic to different sites that deliver the same application or service.
• In the case of failure of the production system, the standby environment will be scaled up for production load, and DNS records will be changed to route all traffic to AWS.	• During failover, you can quickly increase compute capacity by using AWS Auto Scaling or by resizing your instances to a larger size.
• If your data is constantly changing and failover occurs, you would have to reverse replicate your data in the DR site back to the primary site, so that any data updates received while the primary site was down can be replicated back, without the loss of data.	• Multiple services in AWS such as RDS offer a multi-AZ feature which allows you to provision resources in a different location for a more fault-tolerant setup.
	• If your data is constantly changing and failover occurs, you would have to reverse replicate your data in the DR site back to the primary site, so that any data updates received while the primary site was down can be replicated back, without the loss of data.